

The screenshot shows the AVR Studio IDE with the following components:

- Registers Window:** Displays the state of various registers. The 'R' registers (R0-R7) are shown with their values. The 'Sys' register is also visible.
- Assembly View:** Shows the disassembled code for the 'expel6.c' file. The code includes a delay function and a main function that prints 'Hello World' to the UART.
- UART #1 Window:** Shows the output of the program, which is 'Hello World' repeated multiple times.

The screenshot shows the Keil uVision IDE with the following components:

- Registers Window:** Displays the status of various registers. The 'Sys' register is highlighted, showing a value of 0x01.
- Assembly View:** Shows the assembly code for the 'main' function. The code includes a loop that reads characters from the serial port and prints them. The code is as follows:


```

11:  ch = SBUF; // Read received character
12:
13:  printf("a>>> INTERRUPT RECEIVED: ");
14:  printf("%c\n", ch);
15:  }
16:  }
17:
18:  void delay(void)
19:  {
20:  unsigned int i;
21:  for(i = 0; i < 50000; i++); // Delay for visibility
22:  }
23:
24:  void main(void)
25:  {
26:  SOCON = 0x50; // UART mode 1, 8-bit, REN enabled
27:  TRASN = 0x00; // Trans mode 0 (data only)

```
- Command Window:** Shows the output of the program, which is 'Hello World'.